Pre-Workout Ingredients and Label Transparency

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1. Background

The data come from the National Institutes of Health Dietary Supplement Label Database (DSLD) [1], which records information printed on supplement labels. The database includes product details, ingredients, declared amounts, serving information, and label statements. Each product has a DSLD ID, which links the product-level and ingredient-level files. Since the database reports label information, it can be used to study ingredient patterns and dosage disclosure. It does not confirm what is in the product or assess safety, effectiveness, or regulatory compliance.

2. Research Questions

Which ingredients appear most often in pre-workout supplements, and do products with more ingredients tend to disclose a smaller share of individual doses?

3. Method

The three DSLD files were imported into RStudio and cleaned with tidyverse. Five exact duplicate rows were removed, DSLD IDs were converted to the same data type, extra spaces were trimmed, and blank strings were changed to missing values. Conventional nutrition items were excluded so the analysis focused on supplement ingredients. One record was then retained for each DSLD ID-ingredient pair, so an ingredient listed under more than one serving size was counted once. The final ingredient-level dataset contained 5,520 unique label-ingredient records from 394 product labels.

Two measures were created for each label. Ingredient count was used to represent formulation complexity. Strict disclosure rate was calculated as the percentage of unique ingredients with a complete numerical dose. Spearman rank correlation was used to test the relationship because many labels had a disclosure rate of 100%, meaning the values were not normally distributed.

The cleaned ingredient-level and label-level datasets were exported from RStudio to Tableau. Tableau was used to create the common-ingredient chart, disclosure categories, summary measures, and the scatterplot of ingredient count against disclosure rate. The charts were then combined into one dashboard.

4. Results

4.1 Data Source

The raw data were obtained from the NIH Dietary Supplement Label Database (DSLD), which catalogs information printed on dietary supplement labels sold in the United States (NIH Office of Dietary Supplements, n.d.). Three related files—ProductOverview, DietarySupplementFacts, and LabelStatements—were linked by DSLD ID. The raw data and cleansing process can be found at

https://github.com/Ningyu6/Pre-workout-label-anazalysis.git

4.2 Rank of Common Ingredients

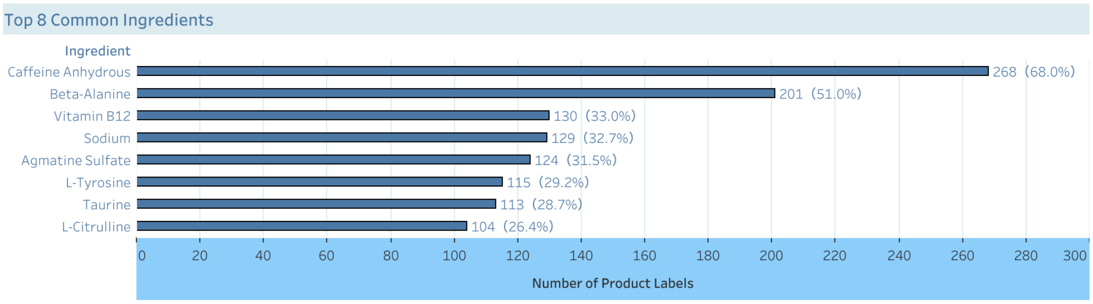

Figure 1 Tableau dashboard of common ingredients

Most products use a small group of recurring ingredients, while the rest of the ingredient list varies more widely. This suggests that many products start from a similar base and differ through secondary ingredients, ingredient forms, and doses. Frequency shows what is common on labels; it does not show which ingredients work best.

4.3 Ingredients Transparency

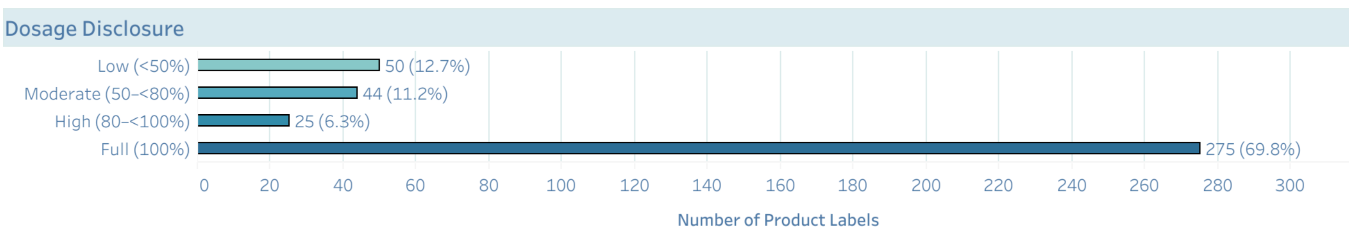

Figure 2 Tableau dashboard of dosage disclosure

Disclosure is clustered at the top of the scale rather than spread evenly across the four groups. Of the 394 labels, 275 reported a dose for every ingredient included in the analysis. The other labels left at least one dose incomplete, which may reflect grouped ingredients or proprietary blends.

4.4 The Relationship between Label Complexity and Dosage Disclosure

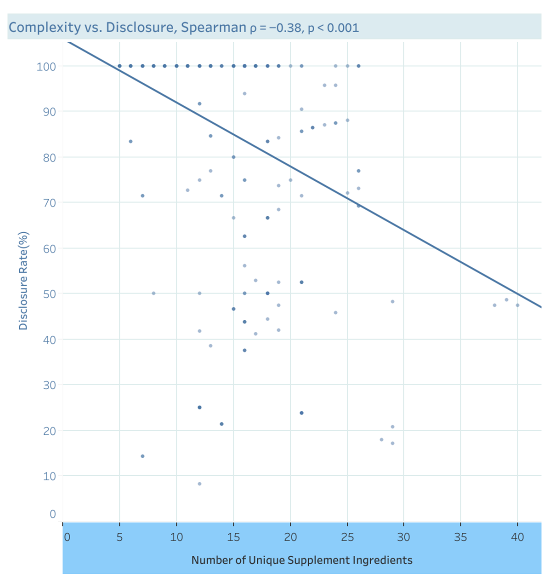

Figure 3 Tableau dashboard of complexity and dosage disclosure

The scatterplot has a downward pattern: labels with more unique ingredients generally disclose a smaller share of doses. Spearman's rho was -0.38 ($p < 0.001$), showing a moderate negative relationship. However, some labels with many ingredients still report every dose, so ingredient count does not explain disclosure by itself.

5. Discussion

Pre-workout products share several common ingredients, but their secondary ingredients and reporting practices vary. Two products can therefore look similar at first glance while differing in the rest of the formula and in how much dosage information they provide.

The negative relationship suggests that complete reporting becomes less common as ingredient lists grow. This does not mean that a complex formula must be less transparent, since many complex labels still disclose every dose. Part of the difference might be affected by factors of proprietary blends and manufacturer reporting choices.

This study is limited to information printed on product labels. It does not test ingredient accuracy, effectiveness, or safety. And the correlation does not show cause and effect.

6. Conclusion

Pre-workout products tend to use a common ingredient base, but they differ in their additional ingredients and dosage disclosure practices. Most labels reported every analyzed dose. However, disclosure was generally lower on labels with more ingredients. Ingredient count helps describe this pattern but does not fully explain it.

7. Reference

[1] National Institutes of Health, Office of Dietary Supplements, “Dietary Supplement Label Database (DSLD).” https://ods.od.nih.gov/Research/Dietary_Supplement_Label_Database/